



The UK AI Skills Hub

A structural design gap — and what the evidence suggests is missing

Status: preliminary findings from an independent research project. The wider project remains at the scoping stage; an initial audience and pilot-unit direction have since been selected for testing (§6). Evidence and inference are distinguished throughout, and every source is tagged by its likely interest. This is not a finished proposal, nor a claim that the underlying problem has been exhaustively verified.

Executive summary

The UK Government's AI Skills Hub — a national platform under Innovate UK and Skills England, expanded in January 2026 to all UK adults with an ambition to upskill 10 million workers by 2030 — has a specific, independently corroborated design concern: it functions as a directory of pre-existing courses organised by technology vendor, not a structured programme mapped to the competencies employers actually need. An LSE academic analysis and two commercial reviewers, testing the platform separately, converge on the same gaps: no competency mapping, no diagnostic-driven sequencing, no embedded practice, no support for organisational readiness. The proposed direction is grounded in existing research rather than invented from scratch: the same government programme that diagnosed the UK's AI skills gap has already published PRIMES, a framework describing evidence-informed training principles, and Finland's Elements of AI shows a coherent, non-directory approach can reach millions without a large budget. Grounded AI Practice, an evidence-led research project, is using this gap to inform its own first practical output — detailed in §6.

1. The platform and the problem

The AI Skills Hub is a national training-navigation platform delivered under the BridgeAI programme, developed by Innovate UK in partnership with Skills England with delivery support from PwC and industry partners. Initially focused on four sectors, it was expanded on 28 January 2026 to all UK adults, with a stated ambition to upskill 10 million workers by 2030 and a reported £4.1 million investment.

The core critique, detailed below, is structural rather than about content quality: the platform curates existing third-party courses but does not sequence them against a competency framework, personalise pathways to a learner's actual level, or provide embedded practice. Reviewers characterise it as “a directory, not a programme.”

2. Independent corroboration

Three separately-authored reviews, none citing the others, converge on overlapping specific findings:



“A mixture of existing training content loosely connected by the topic of AI.”

— LSE Impact of Social Sciences blog (Cardoso-Silva & Brown), on the AI Skills Hub's course organisation

Source	Evidential role	Key finding
LSE Impact of Social Sciences blog (Cardoso-Silva & Brown) Independent academic — LSE Data Science Institute / Digital Skills Lab	Independent policy analysis	Courses organised by technology vendor, not by the competencies the government's own research identifies as most needed (72%+ of surveyed organisations: non-technical skills).
Techosaurus / Quantum Rise Commercial AI-training providers — competing interest (flagged)	Observed platform behaviour	Five flaws: no organisational-readiness support, vendor/platform lock-in, no practice sandbox, no diagnostics or scaffolding, disconnect from how people actually use AI.
The Human Co. Commercial AI-training/consulting — competing interest (flagged)	Observed platform behaviour	Rated the £4.1m platform 3/10 on design despite adequate course content; intermediate learners shown ~71% beginner-level material; no sequencing; progress tracking “nearly invisible.”

Note on interest: two of the three reviewers run commercial AI-training businesses that compete with the platform they reviewed, and are flagged accordingly. Their specific, hands-on structural findings nonetheless align closely with the independent academic critique — the corroboration is across source types, not just repetition of one view. That corroboration covers the broad structural problem; specific figures such as “71% beginner-level material” or the “3/10” rating come from individual reviews, not a replicated evaluation, and should be read as illustrative rather than measured consensus.

3. The underlying cause: “a directory, not a programme”

The pattern across all three reviews is the same: a library of courses of roughly equal prominence, organised by technology or vendor rather than by skill, with no diagnostic step to route learners to their actual level, no mandated sequencing, and no built-in low-stakes practice environment. One reviewer found that intermediate learners were shown roughly 71% beginner-level content — nearly doubling realistic completion time.

The notable irony is that this is not a case of missing evidence. The same government-commissioned research programme (Dr Nisreen Ameen, Royal Holloway, University of London, in partnership with Skills England, drawing on 23 workshops, 10 employer case studies and a 536-response survey) has already published a detailed framework describing evidence-informed principles for more effective AI training — summarised below. No visible competency mapping, diagnostic sequencing or PRIMES alignment was identified in the platform materials the three reviews examined, which supports an inference that the framework was not applied comprehensively to the Hub's design — though none of the three reviewers had access to the Hub's internal design documentation, so this remains an inference from external behaviour, not a confirmed finding.

4. What good looks like: two reference models

4.1 PRIMES — the government's own evidence-based framework



Published 10 June 2026 as part of the same “What Works for AI Upskilling in the UK” research programme, PRIMES sets out six accreditation-style principles for effective AI training. Each row below adds one concrete, checkable requirement from the framework’s own criteria document, not just its six names:

Practical	Links to real tasks and decisions; recognises existing informal AI use rather than assuming a zero starting point; makes explicit when AI should and should not be used.
Reachable	Accessible in time, format and language; explicitly designed around intersecting barriers (income, age, disability, gender), not a single “average learner.”
Integrated	Fits into existing systems and standards; mandatory before staff use AI on regulated, confidential or safety-critical work.
Modular	Short, stackable units (30–90 minutes); learners can enter at different levels and return as needs change.
Expandable	Builds transferable skills usable across tools and roles, not single-vendor competence.
Sustainable	Reviewed on a set cadence (3- and 6-month revisits); tracks confidence and quality of judgement over time, not just enrolment.

Common pitfalls

The same guide names 13 common training pitfalls found across the organisations surveyed. Three are named specifically:

- Training is too fast, too technical, or assumes prior knowledge.
- Existing skills and informal AI use are not recognised.
- Training focuses on tools rather than transferable skills.

Where employers actually are

The same guide’s companion “AI Skills Adoption Pathway” shows survey respondents’ organisations still overwhelmingly concentrated in early stages:

Stage	Share of organisations
Awareness	21%
Exploration	19%
Integration	7%
Strategy	4%
Scaling	1%

Most respondents’ organisations remain concentrated in the earliest two stages (40% combined); only 12% have reached Integration, Strategy or Scaling combined.

4.2 Elements of AI — evidence a coherent alternative can scale in reach

Elements of AI, a free course from MinnaLearn and the University of Helsinki, offers a working counter-example: vendor-neutral, requiring no programming or technical prerequisite, structured as six short chapters combining theory, practical exercises and self-reflection, completed at the learner’s own pace. It has reportedly reached approximately one million learners globally and is



cited by the LSE reviewers as the model the UK's own platform should have followed. This demonstrates reach at scale — it is not, on its own, evidence of measured learning outcomes or workplace capability transfer, which is a different, harder claim than enrolment or completion.

5. Emerging design principles for a credible alternative

Synthesising PRIMES, the platform reviews and the Gradual Release of Responsibility model produces seven provisional design principles for Grounded AI Practice to test, not settled conclusions:

- 1. Modular** — Units of roughly 30–90 minutes with clear entry and re-entry points, rather than long, fixed-length courses.
- 2. Diagnostic-driven** — Sequencing that shifts responsibility from instructor to learner progressively — modelling, then guided practice, then independent practice — rather than jumping straight to unsupported application.
- 3. Practice-led** — Embedded, low-stakes practice (“sandbox” activity), not instructional content alone.
- 4. Bounded** — Explicit guidance on when AI should and should not be used, not only how to use it.
- 5. Transferable** — Tool-agnostic skills prioritised over single-vendor competence.
- 6. Revisitable** — Planned revisit points and outcome tracking (for example, 3- and 6-month check-ins) rather than one-off completion metrics.
- 7. Confidence-calibrating** — Training designed to build calibrated confidence, not just knowledge — see below.

WARNING



Surface knowledge can be riskier than none

Literacy training can improve confidence calibration — trained analysts matched their confidence to their actual accuracy after training (Kamali et al., 2026). But an unverified medRxiv preprint on AI-trained physicians (see Sources) found automation bias persisting even after 20 hours of training in high-stakes clinical use — not treated as settled here. Pair knowledge with deliberate practice judging one's own accuracy, not knowledge alone.

6. Where this sits in the project

Grounded AI Practice's own direction has moved from open scoping to a set of working decisions since this research began:

Initial audience	Individuals seeking practical, everyday AI literacy, with particular attention to employees at small organisations who lack the employer-provided L&D infrastructure that PRIMES and the AI Skills Hub both assume.
First output	A single pilot learning unit — not a roadmap, course library or platform — sized to PRIMES' modular criterion (30–90 minutes) and sequenced using the Gradual Release of Responsibility pattern (§5), tested with real learners before any wider structure is built.



Pilot capability	Effective prompting, working title “Effective prompting — what’s really happening when you hit send” — chosen to show a learner the gap between what they type and what the model actually does with it, while teaching an immediately usable skill.
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None of this claims to have solved the design problem documented above; a single pilot unit is a test of an approach, not proof it works. But each of these decisions was made specifically to avoid the failure mode this briefing documents: one small, coherent, well-sequenced unit — not a library of loosely-connected content — is the more direct test of whether a “programme, not a directory” approach can work at a much smaller scale than the Hub’s £4.1 million platform.

7. A note on method

Findings in this briefing draw on primary sources read directly wherever possible, with every source tagged by its likely interest (government/official, independent/academic, vendor/commercial, or advocacy/membership body) so that the concentration of the evidence base is visible rather than hidden. Where a finding would materially affect the project’s direction, confirming and disconfirming evidence has been sought together rather than assuming the initial hypothesis. A small number of leads referenced during this research (for example, further critical commentary on the AI Skills Hub in the trade press) have not yet been independently verified and are not included as findings here.

This briefing was first drafted 24 July 2026 and updated since with three further findings (PRIMES’ full criteria, a named sequencing model, and confirm/disconfirm evidence on training and overconfidence) logged as the project’s research continued — see RESEARCH_LOG.md Entries 026–028.

Specific limitations of this review, stated directly:

- No access to the Hub’s internal design documentation — findings rest on external behaviour and published material, not internal design intent.
- No learner interviews or usage analytics beyond what the three reviews themselves report.
- No independent replication of the commercial reviewers’ hands-on tests.
- Limited evidence of measured learning outcomes for either the Hub or the comparator models cited.
- The platform’s design may continue to change after this briefing was written.
- This review evaluates the Hub’s structure, not the quality of every individual course it curates.

Sources

Innovate UK / Skills England, UK AI Skills Hub (BridgeAI programme), expanded to all UK adults 28 January 2026. aiskillshub.org.uk

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Kentucky Department of Education, “Evidence-Based Instructional Practices: Explicit Teaching and Modeling” (Gradual Release of Responsibility framework). education.ky.gov

Kamali, Gerstner, Hullman & Groh, “Generative AI Literacy Training Improves Intelligence Analysts’ Discrimination of Real and AI-Generated Images,” arxiv.org/pdf/2606.28510

“Automation Bias in Large Language Model Assisted Diagnostic Reasoning Among AI-Trained Physicians,” medrxiv.org/content/10.1101/2025.08.23.25334280 (preprint, not peer-reviewed; primary source could not be independently verified).

Full source list, interest tagging, and detailed entry-by-entry findings are maintained in `RESEARCH_LOG.md` within the Grounded AI Practice project repository.

